The documentation and process conversion measures necessary to comply with this revision shall be completed by 14 September 1997

INCH POUND

MIL-PRF-19500/398C
14 June 1997
SUPERSEDING
MIL-S-19500/398B
25 January 1995

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-FREQUENCY
TYPES 2N3866, 2N3866A JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN silicon, VHF-UHF amplifier transistor. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (TO-39).

1.3 Maximum ratings.

P_T 1/ T_A = 2/	V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_J and T_{STG}	$R\theta JC$
W	V dc	V dc	V dc	A dc	°C	°C/W max
1.0	60	30	3.5	0.4	-55 to +175	35

1/ Derate linearly 5.71 mW/°C above T_A ≥+25°C.
2/ T_A = Room ambient as defined in the general requirements of MIL-PRF-19500.

1.4 Primary electrical characteristics.

	h_{FE} 1/		h_{fe}		$V_{CE(SAT)}$	C_{obo}	P_{out1}	P_{out2}
	V_{CE} = 5.0 V dc, I_C = 50 mA dc		V_{CE} = 15 V dc, I_C = 50 mA dc, f = 200 MHz		I_C = 100 mA dc, I_B = 10 mA dc	V_{CB} = 28 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	V_{CC} = 28 V dc, P_{in} = 0.15 W, f = 400 MHz	V_{CC} = 28 V dc, Pin = 0.075 W, f = 400 MHz
	2N3866	2N3866A	2N3866	2N3866A				
					V dc	pF	W	W
Min	15	25	2.5	4.0			1.0	0.5
Max	200	200	8.0	7.5	1.0	3.0	2.0	

1/ Pulsed (see 4.5.1)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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2. DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Associated detail specification. The individual item performance requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

η: (eta) Collector efficiency = $\frac{\text{rf power out}}{\text{dc power in}} \times 100$

Pin: Input power

Pout: Output power

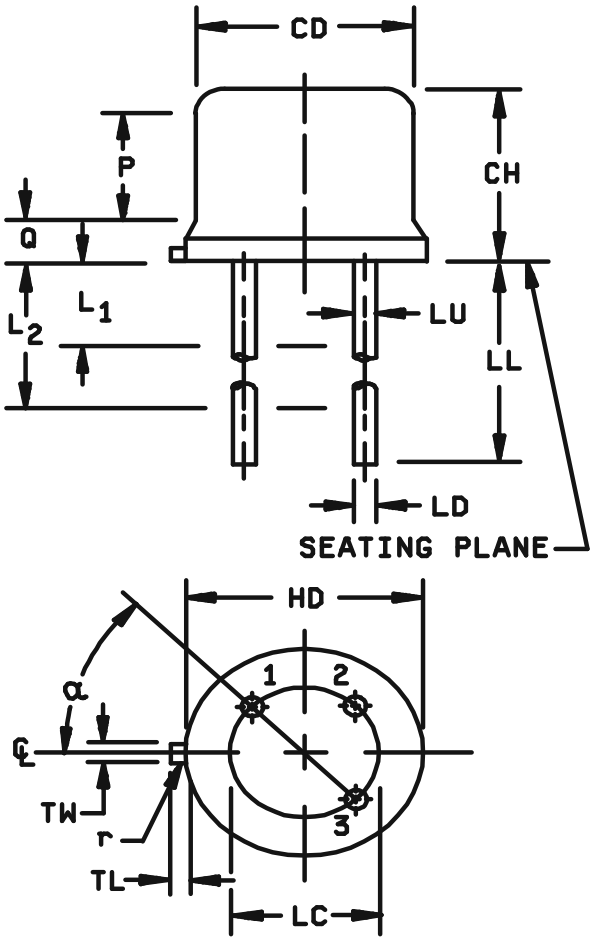
3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions for the purpose of interchangeability shall be as specified on figure 1 herein.

3.3.1 Lead finish. Unless otherwise specified, lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein.

3.3.2 Transistor construction. These devices shall be constructed in a manner and using materials which enable the transistors to meet the applicable requirements of MIL-PRF-19500 and this document.

3.4 Marking. Devices shall be marked as specified in MIL-PRF-19500.

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NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 inch (0.28 mm).
4. TL measured from HD maximum.
5. Outline in this zone is not controlled.
6. CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
7. Leads at gauge plane .054 + .001, - .000 inch (1.37 + 0.03, - 0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedure shown on figure 2.
8. LU applies between L_1 and L_2. LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
9. All three leads.
10. The collector shall be electrically and mechanically connected to the case.
11. r (radius) applies to both inside corners of tab.

FIGURE 1. Physical dimensions.

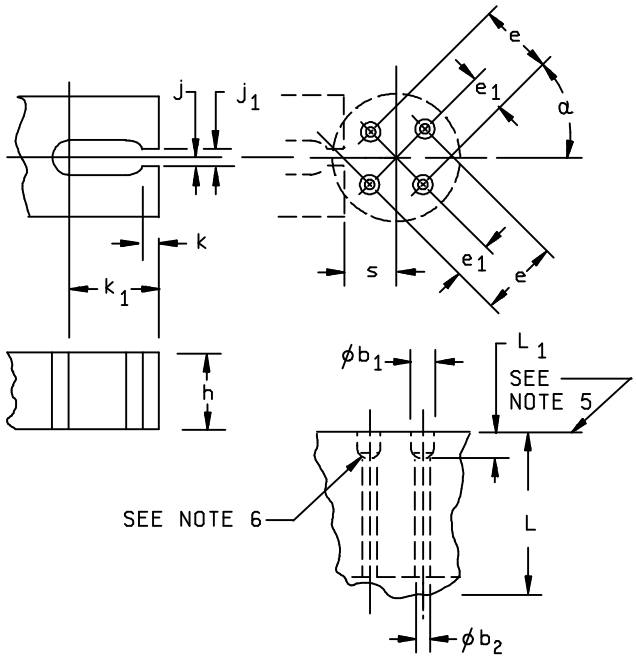
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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CH	.240	.260	6.10	6.60	
LC	.200 TP		5.08 TP		7
LD	.016	.021	0.41	0.53	8, 9
LU	.016	.019	0.41	0.48	8, 9
HD	.335	.370	8.51	9.40	
CD	.305	.335	7.75	8.51	6
TW	.028	.034	.71	.86	3
TL	.029	.045	.74	1.14	4
LL	.500	.750	12.70	19.05	8, 9
L_1		.050		1.27	8, 9
L_2	.250		6.35		8, 9
P	.100		2.54		6
O					5
r		.010		.25	11, 3
α	45° TP		45° TP		7

FIGURE 1. Physical dimensions - Continued.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
$\varnothing b_1$	.0595	.0605	1.51	1.54	
$\varnothing b_2$	.0325	.0335	.83	.85	
e	.1409	.1419	3.58	3.60	
e_1	.0702	.0712	1.78	1.81	
h	.150 Nom		3.81 Nom		
j	.0175	.0180	0.44	0.46	
j_1	.0350	.0355	0.89	0.90	
k	.009	.011	0.23	0.28	
k_1	.125 Nom		3.18 Nom		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
S	.182	.199	4.62	5.05	3
α	44.9°	45.1°	44.9°	45.1°	



NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The location of the tab locator, within the limits of indicated, will be determined by the tab and flange dimensions of the device being checked.
4. The following gauging procedure shall be used: The use of a pin straightener prior to insertion in the gauge is permissible. The device being measured shall be inserted until its seating plane is .125 ±.010 inch (3.18 ±0.25 mm) from the seating surface of the gauge. A spacer may be used to obtain the .125 inch (3.18 mm) distance from the gauge seat prior to force application. A force of 8 ounces ±.50 ounces then be applied parallel and symmetrical to the advice's cylindrical axis. When examined visually after the force application (the force need not be removed), the seating plane of the device shall be seated against the gauge.
5. Gauging plane.
6. Drill angle.

FIGURE 2. Gauge for lead and tab location.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.3 Screening (JANTX level only). Screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of Appendix E, of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	I_{CEO} and h_{FE1}	Not applicable
11	I_{CEO} and h_{FE1}; ΔI_{CEO} = 100 percent of initial value or 2 µA dc, whichever is greater. Δh_{FE1} = ±20 percent of initial value.	I_{CEO} and h_{FE1}
12	See 4.3.1	See 4.3.1
13	ΔI_{CEO} = 100 percent of initial value or 2 µA dc, whichever is greater; Δh_{FE1} = ± 20 percent of initial value; subgroups 2 and 3 of table I herein.	ΔI_{CEO} = 100 percent of initial value or 2 µA dc, whichever is greater; Δh_{FE1} = ± 20 percent of initial value; subgroup 2 of table I herein.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_A = room ambient as defined in the general requirements of MIL-STD-750; V_{CB} = 28 V dc; P_T = 1.0 W.

4.3.2 Thermal impedance ($Z_{\theta JX}$ measurements) for initial qualification or requalification. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101 (read and record value $Z_{\theta JX}$) derived conditions, limits, and thermal response curve shall be supplied to the qualifying activity on the qualification lot prior to qualification approval.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIa (JANS) and table VIb (JANTX and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) and Deltas shall be in accordance with table II, herein.

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4.4.2.1 Group B inspection, appendix E, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	2037	Test condition A.
B4	1037	V_{CB} = 28 V dc; P_T = 1 W at T_A = room ambient as defined in the general requirements of MIL-STD-750, $t_{on} = t_{off}$ = 3 minutes minimum for 2,000 cycles. No heat sink nor forced air on the device shall be permitted.
B5	1027	V_{CB} = 28 V dc; T_A = +125°C ± 25°C for 96 hours; P_T = 1 W at T_A = +100°C or adjusted as required by the chosen T_A to give an average lot T_J = +275°C. Marking legibility requirements shall not apply.

4.4.2.2 Group B inspection, appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	1027	T_A = room ambient as defined in the general requirements of MIL-STD-750, V_{CB} = 28 V dc; P_T = 1 W.
B3	2037	Test condition A.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with table II, herein.

4.4.3.1 Group C inspection, appendix E, table VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	2036	Test condition E.
C6	1026	T_A = room ambient as defined in the general requirements of section 4.5 of MIL-STD-750, V_{CB} = 28 V dc; P_T = 1. W.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in MIL-STD-750.

4.5.2 Thermal resistance (qualification only). Thermal resistance measurements shall be conducted in accordance with test method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power application shall be 79 mA dc minimum.

b. Collector to emitter voltage magnitude shall be 20 V dc minimum.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be +25°C ≤ T_R ≤ +75°C and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to case.

f. Maximum limit of $R_{\theta JC}$ shall be 35°C/W.

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4.5.3 Power-output and collector-efficiency measurements. The device shall be tested in the circuit of figure 3 using the procedure outlined on figure 4. The specified conditions shall be applied and the variable capacitors adjusted to obtain maximum power output. When the maximum power output is obtained, the collector current shall be measured and recorded. The collector efficiency shall be computed as follows:

$$\eta \text{ in percent} = \frac{P_O \text{ (watts)} \times 100}{28 \times I_C \text{ (amperes)}}$$

4.5.4 Burnout by pulsing. The devices shall be tested in the circuit of figure 5. The voltage source shall be increased from zero until the specified current is reached. The current shall be maintained for the specified time.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Collector-emitter breakdown voltage	3011	Bias condition D; I_C = 5 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	30		V dc
Collector-emitter breakdown voltage (unclamped inductive)	3011	V_{BE} = -5 V dc; I_C = 40 mA dc (see figure 6)	$V_{(BR)CEX}$	55		V dc
Collector-base breakdown voltage	3001	Bias condition D; I_C = 100 µA dc; pulsed (see 4.5.1)	$V_{(BR)CBO}$	60		V dc
Emitter-base breakdown voltage	3026	Bias condition D; I_E = 100 µA dc; pulsed (see 4.5.1)	$V_{(BR)EBO}$	3.5		V dc
Collector-emitter cutoff current	3041	Bias condition D; V_{CE} = 28 V dc	I_{CEO}		20	µA dc
Collector-emitter cutoff current	3041	Bias condition C; V_{CE} = 55 V dc	I_{CES1}		100	µA dc
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 50 mA dc; pulsed (see 4.5.1)	h_{FE1}			
2N3866				15	200	
2N3866A				25	200	
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 360 mA dc; pulsed (see 4.5.1)	h_{FE2}			
2N3866				5.0		
2N3866A				8.0		
Collector-emitter saturated voltage	3071	I_C = 100 mA dc; I_B = 10 A dc; pulsed (see 4.5.1)	$V_{CE(sat)}$		1.0	V dc
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition C; V_{CE} = 55 V dc	I_{CES2}		2.0	mA dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Low temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 50 mA dc; pulsed (see 4.5.1)	h_{FE3}			
2N3866				7		
2N3866A				12		
Subgroup 4						
Magnitude of small-signal short-circuit current transfer ratio	3306	V_{CE} = 15 V dc; I_C = 50 mA dc; f = 200 MHz	$\lvert h_{fe} \rvert$			
2N3866				2.5	8.0	
2N3866A				4.0	7.5	
Open circuit output capacitance	3236	V_{CB} = 28 V dc; I_E = 0	C_{obo}		3.0	pF
Power output		V_{CC} = 28 V dc; P_{in} = 0.15 W; f = 400 MHz (see figure 3 and 4.5.3)	P_{1out}	1.0	2.0	W
Power output		V_{CC} = 28 V dc; P_{in} = 0.075 W; f = 400 MHz (see figure 3 and 4.5.3)	P_{2out}	0.5		W
Collector-efficiency		V_{CC} = 28 V dc; P_{in} = 0.15 W; f = 400 MHz (see 4.5.3)	η_1	45		%
Collector-efficiency		V_{CC} = 28 V dc; P_{in} = 0.075 W; f = 400 MHz (see 4.5.3)	η_2	40		%
Subgroups 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Groups B and C electrical and delta measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector-emitter breakdown voltage	3011	Bias condition D; I_C = 5 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	30		V dc
2.	Collector-emitter breakdown voltage (unclamped inductive)	3011	V_{BE} = -5 V dc; I_C = 40 mA dc (see figure 6)	$V_{(BR)CEX}$	55		V dc
3.	Collector-base breakdown voltage	3001	Bias condition D; I_C = 100 µA dc; pulsed (see 4.5.1)	$V_{(BR)CBO}$	60		V dc
4.	Emitter-base breakdown voltage	3026	Bias condition D; I_E = 100 µA dc; pulsed (see 4.5.1)	$V_{(BR)EBO}$	3.5		V dc
5.	Collector-emitter cutoff current	3041	Bias condition D; V_{CE} = 28 V dc	I_{CEO}		20	µA dc
6.	Collector-emitter cutoff current	3041	Bias condition C; V_{CE} = 55 V dc	I_{CES1}		100	µA dc
7.	Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc I_C = 50 mA dc pulsed (see 4.5.1)	h_{FE1}			
	2N3866				15	200	
	2N3866A				25	200	
8.	Collector-emitter saturated voltage	3071	I_C = 100 mA dc; I_B = 10 A dc; Pulsed (see 4.5.1)	$V_{CE(sat)}$		1.0	V dc
9.	Collector-emitter (voltage saturated)	3071	I_C = 100 mA dc; I_B = 10 mA dc; pulsed (see 4.5.1)	$\Delta V_{CE(SAT)}$ 4/	± 50 mV dc change from previous measured value.		
10.	Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 50 mA dc; pulsed (see 4.5.1)	Δh_{FE} 4/	± 25 percent change from initial reading		
11.	Collector-emitter cutoff current	3041	Bias condition D; V_{CE} = 28 V dc	Δ_{CEO} 4/	100 percent or ± 2 µA dc, whichever is greater		

See footnotes on top of next page.

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TABLE II. Groups B and C electrical and delta measurements - Continued.

1/ The electrical measurements for appendix E, table VIa (JANS) of MIL-PRF-19500 are as follows:

a. Subgroup 4, see table II herein, steps 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, and 11.

b. Subgroup 5, see table II herein, steps 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, and 11.

2/ The electrical measurements for table VIb (JANTX and JANTXV) of MIL-PRF-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1, 2, 3, 4, 5, 6, 7, 8, and 11.

b. Subgroup 6, see table II herein, steps 1, 2, 3, 4, 5, 6, 7, 8, and 11.

3/ The electrical measurements for table VII of MIL-PRF-19500 are as follows:

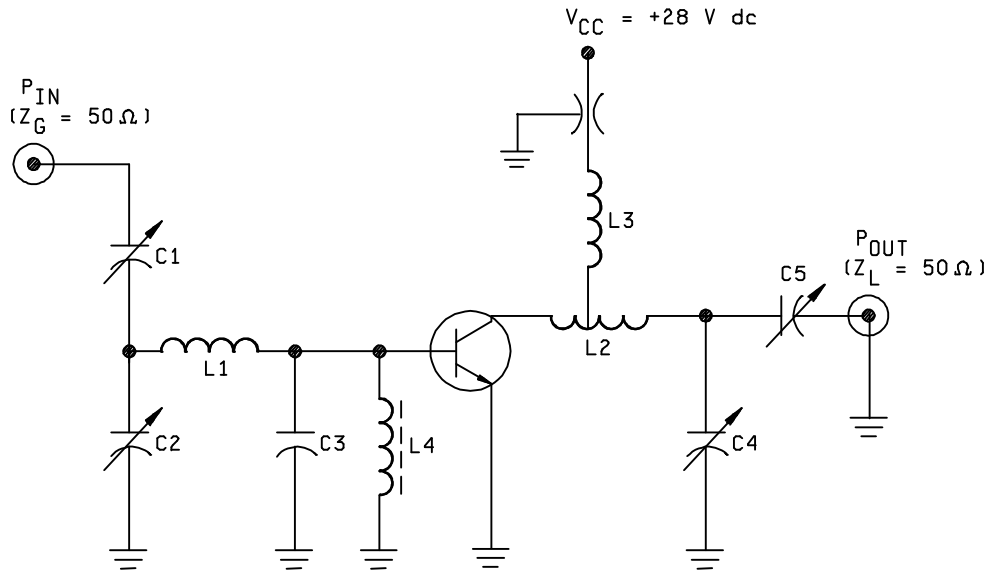
a. Subgroup 2, see table II herein, steps 1, 2, 3, 4, 5, 6, 7, 8, and 11 (JANS).

b. Subgroup 3, see table II herein, steps 1, 2, 3, 4, 5, 6, 7, 8, and 11 (JANS).

c. Subgroup 6, see table II herein, steps 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, and 11 (JANS).

4/ Devices which exceed the group A limits for this test shall not be acceptable.

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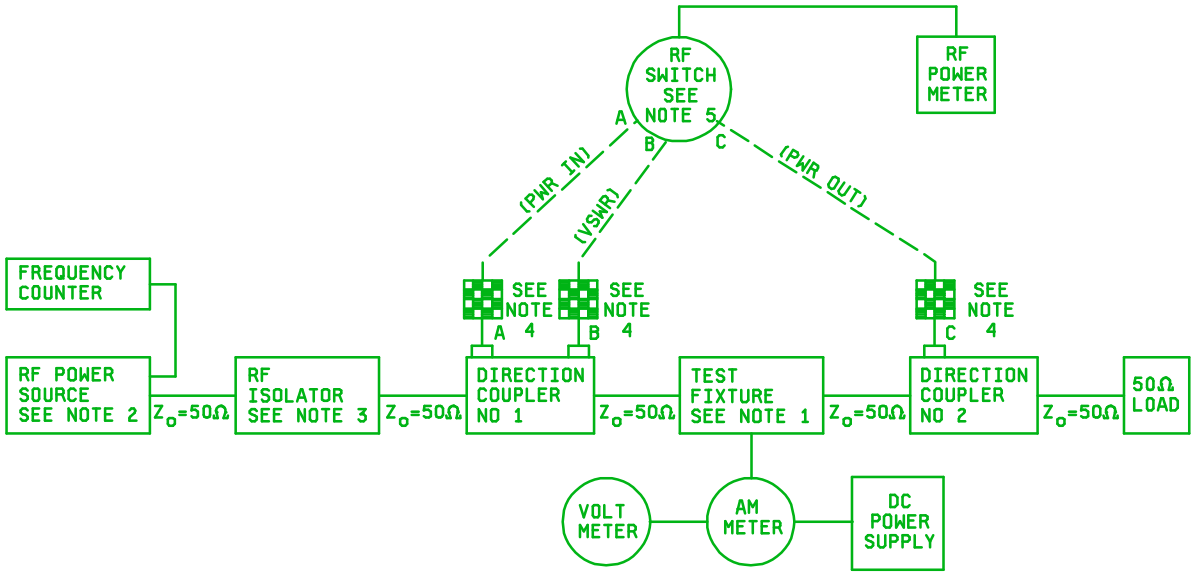


C_1, C_2, C_5 = 3 - 35 pF
C_3 = 24 pF (see note)
C_4 = 0.4 - 7 pF
L_1 = Straight piece number 16 bare tin wire, .625 inch long
L_2 = 3 turns number 16 wire, .250 inch ID, .312 inch long
L_3 = 1 turns number 18 wire, .250 inch ID, .022 inch long
L_4 = Ferrite RF choke, Z = 450 Ω

NOTE: For optimum performance, C_3 should be mounted as close as possible to base lead.

FIGURE 3. Power - output test circuit (400 MHz).

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NOTES:

1. Test fixture is the circuit as described on figure 3.
2. RF power source may be any unit capable of generating desired power level at desired frequency with a harmonic and spurious content at least 20 dB below operating frequency level.
3. The RF isolator may be any device (pad, circulator, ect.) capable of establishing at least 20 dB of isolation (RL > 20 dB) between RF source and test fixture.
4. Variable attenuators (or fixed if calibrated): Attenuator on directional coupler number 2 shall be calculated against known working standard either by means of calibration chart or suitable adjustment if variable. Attenuator at position "A" of directional coupler number 1 shall be calibrated or adjusted so that actual power at test fixture is known. Attenuator at position "B" shall be adjusted to establish sensitivity needed to measure VSWR.
5. RF switch may be eliminated if additional power meters are used.

PROCEDURE:

a. Remove "test fixture" and install jumper between directional coupler number 1 and directional coupler number 2.
b. Set the RF switch to power output position "C".
c. Adjust frequency and power of RF source, as required by specification, and monitor frequency counter and RF power meter respectively (see note 4).
d. Set the RF switch to position "A" and adjust variable attenuator to obtain identical reading as power out in position "C" (see note 4).
e. Reconnect "test fixture" in test setup and insert device.
f. Adjust power supply to 28 V dc.
g. Adjust circuit output tuning for maximum power gain and circuit input tuning for maximum VSWR. (Switch between power in; VSWR, and power out while tuning and repeat as many times as necessary to obtain minimum VSWR and maximum power out. Check power in level before taking final reading. Minimum VSWR is defined as minimum reading obtained on power meter with switch in position "B" and maintaining power in.)

FIGURE 4. RF power output (P_{OUT}) test procedure.

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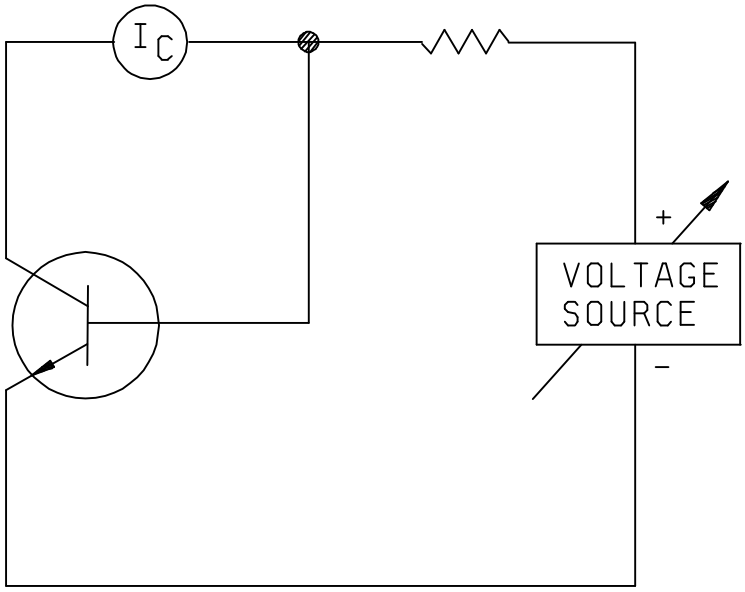
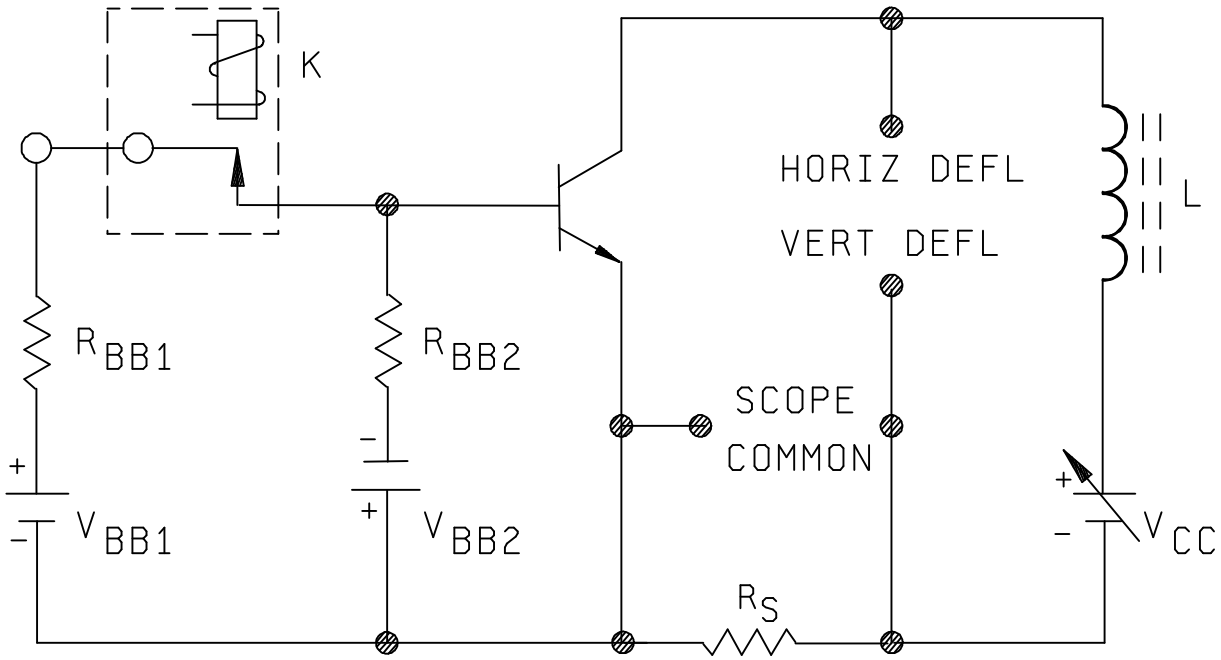


FIGURE 5. Burnout by pulsing test circuit.

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R_{BB1} = 150 Ω.
V_{BB1} = 20 V dc.
K = S.p.s.t. relay, 6 V ac coil (Clare Mercury Relay, model number HGP-1400, or equivalent).
R_{BB2} = 33 Ω.
V_{BB2} = 1.5 V dc
R_S = 1 Ω ± 1 percent, 1/2-watt (noninductive).
V_{CC} = The voltage should be adjusted to approximately 17 volts.
L = 25 mH, 100 mA, 83 Ω dc resistive (Miller number 957, or equivalent).

FIGURE 6. $V_{BR(CEX)}$ (unclamped inductive) test circuit.

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, 3990 E. Broad Street, Columbus OH 43216-5000.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 85, 99

Preparing activity:
DLA - CC

(Project 5961-1853)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/398C	**2. DOCUMENT DATE (YYMMDD)** **970614**

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-FREQUENCY TYPES 2N3866, 2N3866A JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial (614)692-0510 (2) AUTOVON 850-0510
c. ADDRESS (Include Zip Code) Defense Supply Center Columbus, DSCC-VAT, 3990 E. Broad Street, Columbus OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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